

Solution Requirements

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system provides a prediction interface without mandatory user login. Authentication can be added in future versions.</i>	Low
2	Authorization levels	<i>All users can access the crop prediction feature. No separate user roles are implemented in the current version.</i>	Low
3	External interfaces	<i>The React frontend communicates with the Flask REST API, which loads the trained Random Forest model to generate predictions.</i>	High

4	Transactions processing	<i>The system accepts soil and environmental parameters, validates the input, processes the prediction request, and returns the recommended crop.</i>	High
5	Reporting	<i>The application displays the predicted crop recommendation instantly through the web interface.</i>	Medium
6	Business rules, etc.	<i>Crop prediction is based on Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall values using the trained machine learning model.</i>	High
7	Compliance to laws or regulations	<i>The application processes only user-provided agricultural data and does not collect personal or sensitive information.</i>	Medium
8	Other	<i>Input validation ensures that only valid numerical values are accepted before prediction.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	<i>The application should generate crop recommendations quickly after user input.</i>	<i>Response time less than 2 seconds</i>
2	Scalability	<i>The system should support future expansion with additional crops, datasets, and users.</i>	<i>Easily scalable for future enhancements</i>
3	Security & Data Privacy	<i>User inputs are validated before processing, and no personal information is stored.</i>	<i>Input validation for every request</i>
4	Reliability & Availability	<i>The Flask backend should consistently process prediction requests without failure.</i>	<i>99% availability during normal operation</i>
5	Usability & Accessibility	<i>The interface should be simple, responsive, and easy for users to understand on desktop and mobile devices.</i>	<i>Responsive and user-friendly interface</i>
6	Other	<i>The project should be modular and easy to maintain for future improvements.</i>	Modular React and Flask codebase